

Diverting Hydrographs

You can divert any hydrograph into two separate hydrographs. This situation can occur when channels divide or when pond outlet structures are designed to direct the outflow in two different directions. For example, you might want the outflows from an emergency spillway to be separated into one hydrograph and the remaining flows into another hydrograph. You may also want to separate the exfiltration component as a hydrograph.

Hydraflow Hydrographs Extension offers four diversion methods:

- **Divide by constant Q**

Use this method to divide the hydrographs by a constant flow (Q). For example, if you entered 35.5, Hydraflow Hydrographs Extension would generate one hydrograph consisting of all Qs above 35.5 cfs, and another hydrograph that consisted of all Qs equal to or below 35.5.

- **Divide by flow ratio**

Use this method to divide the inflow hydrograph by a ratio. For example, if you entered 0.25, Hydraflow Hydrographs Extension would generate one hydrograph with flows equal to 25% of the inflow ordinates and another hydrograph that consisted of the remaining 75% flow ordinates.

- **Divide by first flush volume**

Use this method if you need to strip off an initial volume from the inflow hydrograph. For example, if your detention pond contains a fore bay, you could extract that volume from the inflow hydrograph first. Set it aside and route the remaining hydrograph through the pond.

- **Divide by pond structure**

Use this method in situations where the inflow hydrograph is a Reservoir Route hydrograph. Then you have the option of dividing the hydrograph by one of the outlet structures for the pond. For example, if the pond has a culvert and a weir structure and you want to redirect the weir flows offsite, you can select the weir structure as the divide. Hydraflow Hydrographs Extension will then create two hydrographs: one consisting of the weir flows, and the other made from the remaining flows.

Use this option after you performed the routing. Using the hydrograph for the pond outflow and the stage-discharge curve, Hydraflow Hydrographs Extension calculates the component flows of the specified outlet structure.

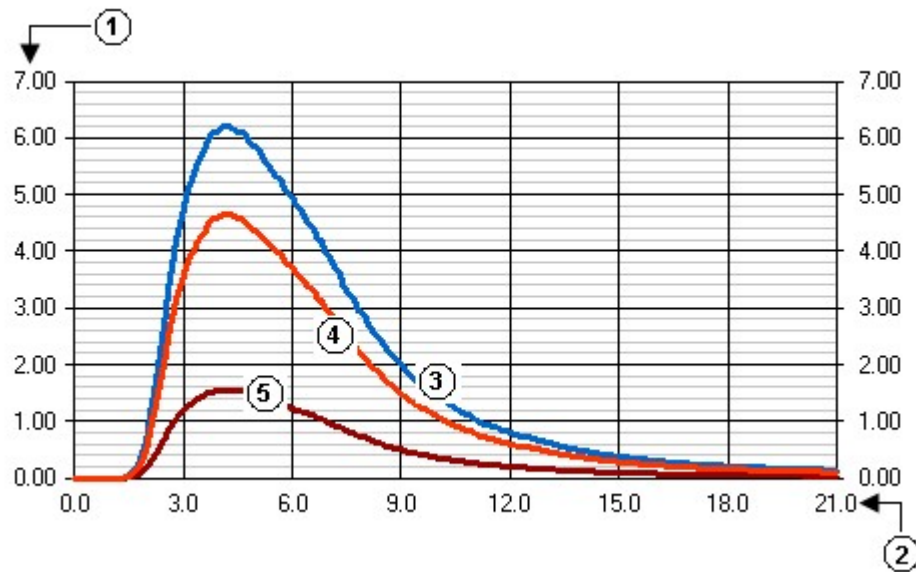
When diverting, Hydraflow Hydrographs Extension creates two hydrographs and labels them as Diversion1 and Diversion2.

To divert a hydrograph

1. In the Model or Hydrograph view, select a hydrograph that you want to to divert.
2. On the toolbox, click  (Divide Hydrograph in Two).
3. In the Divert Hydrographs dialog box, enter the required data.

4. Click OK to return to the initial view, or click Results to view the resulting hydrograph.

The following illustration shows an inflow hydrograph (3) split by a ratio of 75%.



(1) Q (cfs)	(3) Inflow hydrograph	(5) Split hydrograph (25%)
(2) Time (min)	(4) Split hydrograph (75%)	

See Also

- [Divert Hydrographs Dialog Box](#)

Topics in this section

- [Divert Hydrographs Dialog Box](#)

Use this dialog box to enter data for diverting a hydrograph.